

4.3.3 Handling Missing Values (NaN)

خالی خانے NaN ہیں۔

CS XII

Chapter 4: Data and Analysis

Lecture 05 of 9

★ Golden

4.3.3 Handling Missing Values (NaN)

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LEARN IT

Missing cells are **NaN** ("Not a Number"). Statistics crash or lie if you ignore them. Three tools: `isnull()` (True where empty), `fillna(value)` (replace), `dropna()` (delete the row).

```
import pandas as pd
df = pd.DataFrame({
    "Name": ["Ali", "Sara", "Ahmed", "Ayesha"],
    "Marks": [85.0, None, 78.0, 92.0],
    "Grade": ["A", "A", None, "A"],
})
print(df.isnull().sum())
```

OUTPUT

```
Name      0
Marks     1
Grade     1
dtype: int64
```

WORKED EXAMPLE

Replace Marks with the mean of the known marks, and Grade with a label. Known marks 85, 78, 92 → mean = $255/3 = 85.0$.

```
avg = df["Marks"].mean()          # 85.0
df["Marks"] = df["Marks"].fillna(avg)
df["Grade"] = df["Grade"].fillna("Not Assigned")
print(df)
```

OUTPUT

	Name	Marks	Grade
0	Ali	85.0	A
1	Sara	85.0	A
2	Ahmed	78.0	Not Assigned
3	Ayesha	92.0	A

`df = df.dropna()` would instead delete Sara's row and Ahmed's row (any row that still had a NaN).

EXAM TIP

State: identify with `isnull / sum`, fill numeric gaps with the mean, fill categories with a label, or drop rows if filling would invent a lie. Never analyse first and clean second.

COMMON MISTAKE

Filling a salary gap with the mean of a company that includes the CEO can hide poverty (the same outlier problem as the mean). Know *why* the value is missing before you fill.

اردو میں سمجھیں

خالی خانے NaN ہیں۔ isnull تلاش، fillna بھرنا (اکثر اوسط سے)، dropna قطار ہٹانا۔ پہلے صاف کرو، پھر تجزیہ۔

CHECK YOURSELF

1. What does `df.isnull().sum()` return?
2. Does `dropna()` delete a column or a row by default?

Answers: 1. Count of NaNs in each column. 2. Any **row** that contains at least one NaN.

Next lecture: 4.4 Data Organization and Representation